

Decision Integrity Assessment: SPARK Workflow Pipeline

Where the system assists, where it frames, and how to make the boundary visible

Codebase analyzed:	spark-workflow-main (Release 1, March 2026)
Challenge:	3 — “Safe and Stable!”
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1. What SPARK Does Well

This assessment is based on a full read of the SPARK Release 1 codebase. Before identifying gaps, it is important to acknowledge what the system already does right — because the gaps are not design failures. They are the natural next step after a strong foundation.

✓ Three-stage plausibility verification

Risk Screener → Context Checker → Conflict Verifier. The verifier can lower ratings but never raise them. This is a sound architectural decision that prevents escalation bias.

✓ Explicit false-positive prevention

The `context_checker_prompts.py` contains nine explicit rules against false positives (Rules 0–9), including: no semantic escalation, grounding requirement for high ratings, explicit marker requirement for contradiction claims, and strict entity/scope/level matching. This level of prompt discipline is rare.

✓ LLM summary provenance marking

LLM-generated summaries of images/tables are explicitly marked and treated as “weak hints” in the evaluation prompts. High ratings require primary text evidence.

✓ Prompt injection defense

External data is sanitized, tagged, and sandboxed. Injection patterns are stripped. Anti-injection preambles wrap all system prompts.

2. Four Structural Observations

The following observations are not vulnerabilities. They are structural properties of the pipeline that become governance-relevant under operational conditions — specifically at scale, under time pressure, and with Release 2 capabilities.

Observation 1: Rating is a normative judgment presented as neutral assessment

The Risk Screener system prompt states: “keine rechtlichen, fachplanerischen oder normativen Entscheidungen.” But it assigns ratings from 0–100 that determine which claim pairs receive further review. A rating is an interpretive choice: it selects which contradictions matter and which do not.

The Sachbearbeiter sees: “Rating 78 — prüfen.” The Sachbearbeiter does not see: which interpretation the system chose, which alternatives existed, or why this pair was rated higher than another.

Code reference: `risk_screener_prompts.py`, lines 1-50

Observation 2: No human breakpoint between pipeline stages

The FVP workflow in `fvp_workflow.py` runs: Inhaltsextraktion → Formale Prüfung + Inhaltsverzeichnis-Matching (parallel) → Plausibilitätsprüfung. The human receives only the final output. There is no intermediate checkpoint where a Sachbearbeiter can verify extraction quality before it feeds into the plausibility layer.

Consequence: an extraction error (misread number, misclassified document, incorrectly summarized table) becomes a “contradiction” in the plausibility report. The Sachbearbeiter sees a conflict finding, not an extraction artifact.

Code reference: `fvp_workflow.py`, lines 31-89

Observation 3: LLM summary provenance lost in vector layer

The prompts correctly distinguish between original text and LLM-generated summaries. But in the Qdrant vector store, both are stored as embeddings. When the plausibility check retrieves claim pairs for comparison, the provenance distinction may not survive the retrieval step. The Sachbearbeiter cannot tell which plausibility findings are grounded in original documents versus LLM-generated content.

Code reference: `qdrant_wf/workflow.py`; `check_logic_wf/workflow.py`

Observation 4: Release 2 will introduce the critical decision loop

The roadmap (`docs/roadmap.md`) shows Release 2 adds: “Legal review and assessment” and “Drafting of decisions.” When the system drafts the Beschluss, the Sachbearbeiter reviews a pre-written decision document. At high volume, review becomes signature. The system’s framing becomes the decision — not because the Sachbearbeiter is negligent, but because the system’s output looks correct.

This is a documented pattern. In the US, Cigna’s PxDx system processed 300,000 claim denials in two months at 1.2 seconds per review. A federal court found this constituted an “abuse of discretion” (*Kisting-Leung v. Cigna*, March 2025).

3. Proposal: Framing Transparency Layer

The proposal is not to redesign the pipeline. The pipeline is well-built. The proposal is to add a transparency layer that makes visible where the system assists and where it frames — so that the Sachbearbeiter can distinguish between extraction (factual) and interpretation (normative) at every step.

→ Provenance tag per plausibility finding

Each conflict finding in the output report carries a provenance marker: “Based on original document text” vs. “Based on LLM-generated summary of image/table.” This already exists in the prompts. It needs to survive into the output layer.

→ Interpretation disclosure per rating

Each rating above a threshold (e.g., 50) includes a one-line disclosure: “This rating assumes [same entity / same scope / same regulatory reference]. If this assumption does not hold, the rating is not applicable.” The Risk Screener already generates this in the “reasoning” field. It needs to be surfaced to the Sachbearbeiter, not hidden in the JSON.

→ Pipeline stage confidence indicator

The output report includes a per-finding indicator of which pipeline stage generated the finding and what confidence level each stage assigned. This allows the Sachbearbeiter to see: “This contradiction was found by the Risk Screener (rating 72), confirmed by the Context Checker (rating 58), and reduced by the Verifier (rating 45).” The full verification chain becomes visible.

→ Decision boundary logging for Release 2

When “Drafting of decisions” is added in Release 2, each generated Beschluss should carry an explicit marker: “The following elements were generated by the system. The following elements require Sachbearbeiter determination.” This prevents the gradual compression of human oversight under scale.

4. Implementation Sketch (2-Day Hackathon Scope)

Component	What	Where in codebase	Effort
Provenance tag	Carry LLM-summary flag from Qdrant through check_logic to output JSON	qdrant_wf + check_logic_wf + cluster_summarizer	Day 1
Reasoning surfacing	Include Risk Screener reasoning field in final output report	cluster_summarizer_prompts.py + output_schemas	Day 1
Verification chain	Log rating at each stage (screener → checker → verifier) in output	check_logic_wf/workflow.py + output_schemas.py	Day 2
Decision boundary spec	Define output schema for Release 2 Beschluss with system/human markers	New: decision_boundary_schema.py	Day 2

5. The One Question

SPARK is designed as assistance that does not replace the Sachbearbeiter’s decision responsibility.

The question is: **How do we ensure that the Sachbearbeiter can still tell where the system’s framing ends and their own judgment begins?**

Not today, when volumes are low and the system is new.

But in three years, when the system has processed thousands of applications, and the Sachbearbeiter has learned to trust it.

About the author: Marko A. E. Chalupa is the developer of the Decision Integrity Protocol (DIP) and founder of SnapOS Foundation. The DIP Audit Series has published eight governance assessments of real-world automated decision systems, including fraud detection (JPMorgan Zelle), medical necessity determinations (UnitedHealth, Cigna), AML monitoring (TD Bank), EU AI Act enforcement, and autonomous research systems. All publications are available with DOI on Zenodo (ORCID: 0009-0000-6493-4599).

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